

Pixels, Voxels, Contrast and Resolution

Imaging:

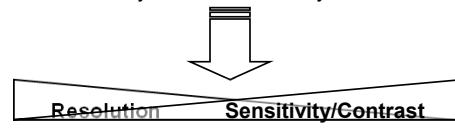
Localized measurement of a **contrast generating** biophysical effect in body/organ of living system

- **What is measured:**
 - Pixels
 - Voxels
 - Noise – affect Contrast
- **2D Image=nxm matrix of pixels**
Pixel = picture element
Image **stack** = set of 2D images representing a volume.
- **3D image=kxnxm matrix of voxels**
Voxel = volume element.

What is Contrast ?

- *Contrast* = difference in signal between tissues one wishes to distinguish or 'resolve', in a given image.
- Contrast-to-noise – Ability to distinguish tissue features against noise.

Increased resolution comes at the price of decreased contrast and therefore sensitivity to detection of tiny features:



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Signal-to-noise v/s Contrast-to-noise

- Signal-to-noise ratio (SNR)
- SNR provides a means to estimate the precision with which the signal S is measured
 - S: signal (or measurement variable)
 - σ : standard deviation of its measurement

$$SNR = \frac{S}{\sigma}$$

When is it possible to have excellent SNR but poor CNR ..?

To discriminate two signals S_1 and S_2 we need more than just good signal to noise ratio. The ability to discriminate the two is assessed using the contrast to noise ratio...

Contrast-to-noise ratio (CNR)

S_1 and S_2 : two signals (or measurement variable) of two different tissues,

σ : standard deviation of their independent measurement (assumed here to be identical and statistically independent)

$$CNR = \frac{S_1 - S_2}{\sigma}$$

CNR provides a means to estimate the precision with which S_1 can be discriminated from S_2

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Optimizing SNR

- It is possible to optimize SNR by performing N repeated measurements S_i .
- Eg: 4 measurements improve the precision by two-fold.
- The precision of the average $\langle S \rangle = \sum S_i / N$ depends on the square root law

$$\langle \varepsilon_i \varepsilon_j \rangle = 0, i \neq j$$

$$\Delta S^2 = \frac{\left(\sum \varepsilon_i^2 \right)}{N^2} + \frac{\left(\sum_{i \neq j} \varepsilon_i \varepsilon_j \right)}{N^2}$$

$$S_i = S + \varepsilon_i$$

$$\text{where } \langle \varepsilon_i^2 \rangle = \sigma^2, \langle \varepsilon_i \rangle = 0.$$

$$\langle \Delta S^2 \rangle = \frac{\sum \langle \varepsilon_i^2 \rangle}{N^2} = \frac{N \sigma^2}{N^2} = \frac{\sigma^2}{N}$$

If S is the true signal (unknown):

$$\langle S \rangle = \sum S_i / N = S + \sum \varepsilon_i / N$$

$$\langle \Delta S \rangle = \frac{\sigma}{\sqrt{N}}$$

$$\Delta S \equiv \langle S \rangle - S = \frac{\sum \varepsilon_i}{N} \quad \Delta S^2 = \frac{\left(\sum \varepsilon_i \right)^2}{N^2}$$

The Problem..?

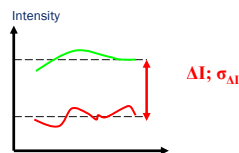
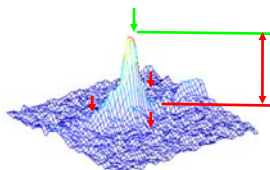
- Results in increased measurement time.

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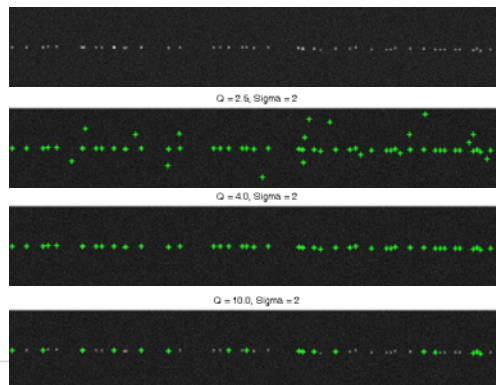
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Statistical Selection of Features



$$I_{max} - I_{BG} \geq Q \cdot \sigma_{\Delta I} ?$$

Q: selection quantile



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Optimizing CNR ?

•Optimizing contrast = choice of experimental parameters (e.g. protocol) to maximize the difference in two tissue signals S_1 and S_2 .
 complex and empirical procedure
 some effects can be predicted/calculated, if the signal behavior can be modeled.

$$\begin{aligned}\frac{d}{dt}(-S_0 t e^{-kt}) &= 0 \\ &= -S_0 e^{-kt} + S_0 k t e^{-kt} \\ &= -S_0 e^{-kt} (1 - kt) = 0 \\ &\quad \boxed{t_0 = 1/k}\end{aligned}$$

Error propagation calculation

Let the signal S be a function $S(k, t)$

k is a tissue property (signal decay rate)

t an experimental parameter (such as time).

Approach:

1. Determine dS/dk
2. Find t_0 where dS/dk is maximal by taking derivative rel. to t

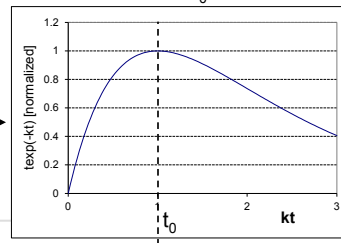
For an exponentially decaying signal, the optimal time of measurement is equal to 1/decay rate

How critical is the choice of t_0 ?

Example: $S(k, t) = S_0 e^{-kt}$

$$\frac{dS(k, t)}{dk} = -S_0 t e^{-kt}$$

Maximum is where derivative with respect to t is zero



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